

Saturated FIML Information, MLR, and FMG under Non-normal Incomplete Data

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Purpose

The useful question is not whether FIML under non-normality has a “Gamma matrix”. It does. The useful question is which curvature and which reference law are being used.

For a FIML goodness-of-fit correction there are three separate objects:

1. the saturated observed-data normal curvature V ;
2. the saturated sandwich covariance $\Gamma_{\text{mis}} = H^{-1} J H^{-1}$;
3. the reference-law reduction of $U\Gamma_{\text{mis}}$, either a scalar mean scaling or the full eigenvalue mixture.

The MCAR simulations should be understood mainly through the third object. Under a correct null and MCAR, the saturated and structured definitions of V have the same asymptotic target. Non-normality changes Γ_{mis} and spreads the eigenvalues of $U\Gamma_{\text{mis}}$; it does not by itself create a different saturated-versus-structured V target.

Observed-data normal H1 model

Let row i have observed index set $o_i \subset \{1, \dots, p\}$. The working FIML H1 model is a normal model for the observed subvector,

$$\ell_i(\eta) = \log \phi_{|o_i|}\{x_{i,o_i}; \mu_{o_i}, \Sigma_{o_i o_i}\}, \quad \eta = (\mu, \text{vech}(\Sigma)).$$

The saturated H1 estimator $\hat{\eta}$ maximizes $\sum_i \ell_i(\eta)$. This is a normal quasi-likelihood when the data are non-normal; the general sandwich theory is the usual misspecified-likelihood theory of White [9].

Define the saturated H1 score

$$h_i(\eta) = \frac{\partial \ell_i(\eta)}{\partial \eta}$$

and let $\eta^\dagger$ denote the population maximizer of $\mathbb{E}\{\ell_i(\eta)\}$. The population bread and meat are

$$H_\eta = -\mathbb{E} \left[\frac{\partial h_i(\eta^\dagger)}{\partial \eta'} \right], \quad J_\eta = \mathbb{E}\{h_i(\eta^\dagger)h_i(\eta^\dagger)'\}.$$

Then

$$\text{avar}\{\sqrt{N}(\hat{\eta} - \eta^\dagger)\} = \Gamma_{\text{mis}} = H_\eta^{-1} J_\eta H_\eta^{-1}.$$

This is the missing-data analogue of the complete-data ADF covariance of sample moments [1]. If the observed-data normal model is correctly specified, information equality gives $J_\eta = H_\eta$. Under non-normality, J_η generally contains the relevant third- and fourth-moment information and is not equal to H_η .

The FIML likelihood-ratio statistic

The restricted SEM is the null $H_0 : \eta = \eta(\theta)$ nested in the saturated H1 model. Let $\hat{\eta}$ be the saturated FIML maximizer and $\eta(\hat{\theta})$ the restricted FIML maximizer of the same observed-data normal log-likelihood. The goodness-of-fit statistic is the ordinary likelihood-ratio,

$$T = 2 \left\{ \sum_i \ell_i(\hat{\eta}) - \sum_i \ell_i(\eta(\hat{\theta})) \right\}, \quad \text{df} = \dim(\eta) - \dim(\theta),$$

i.e. lavaan's default `missing = "ml"` test (the same statistic the nested configural-versus-metric comparison forms as a difference of two such fits). Under a correctly specified normal model with ignorable missingness the information equality $J_\eta = H_\eta$ holds, so $U(V)\Gamma_{\text{mis}}$ is a projector, all nonzero $\lambda_j = 1$, and

$$T \Rightarrow \chi_{\text{df}}^2.$$

This is the only regime in which the unscaled T is asymptotically pivotal. Under non-normality the working normal likelihood is misspecified, information equality fails, and T is no longer χ_{df}^2 ; its limit is the weighted mixture derived below, with mean $\sum_j \lambda_j = \text{tr}\{U(V)\Gamma_{\text{mis}}\}$ rather than df. Every robust correction (MLR, SB, the FMG family) is a different way of referencing T to that mixture instead of to χ_{df}^2 .

Definition of V

The matrix V is the local curvature used to approximate the H1 likelihood in saturated moment coordinates. In the saturated FIML FMG construction,

$$V = H_\eta(\eta^\dagger)$$

in population notation. In the finite-sample magmaan implementation,

$$V_N = - \sum_{i=1}^N \frac{\partial^2 \ell_i(\hat{\eta})}{\partial \eta \partial \eta'}$$

and

$$\hat{\Gamma}_{\text{mis}} = V_N^{-1} \left\{ \sum_{i=1}^N h_i(\hat{\eta}) h_i(\hat{\eta})' \right\} V_N^{-1}.$$

This finite-sample $\hat{\Gamma}_{\text{mis}}$ carries $1/N$ units, while V_N carries N units; their product has the same limiting eigenvalues as the population convention above.

Let the restricted SEM imply saturated coordinates $\eta(\theta)$, and let

$$\Delta = \frac{\partial \eta(\theta)}{\partial \theta'}$$

at the null value or at $\hat{\theta}$. With equality constraints, Δ is first projected into the local free-parameter tangent space. The residual operator for a chosen curvature V is

$$U(V) = V - V\Delta(\Delta'V\Delta)^{-1}\Delta'V.$$

The robust reference law for the FIML LRT is the weighted mixture

$$T \Rightarrow \sum_{j=1}^{\text{df}} \lambda_j \chi_{1j}^2, \quad \lambda_j = \text{eig}_j\{U(V)\Gamma_{\text{mis}}\},$$

the same Satorra–Bentler $U\Gamma$ structure used for robust moment structure tests [7, 6]. When the normal H1 model is correct, $\Gamma_{\text{mis}} = V^{-1}$, so $U(V)\Gamma_{\text{mis}}$ is a rank-df projector and the nonzero eigenvalues are all one.

What MCAR buys

Under MCAR, the missingness pattern is independent of the data values. Suppose the full data have finite second moments (μ_0, Σ_0) . For any observed pattern o ,

$$\mathbb{E}(X_o - \mu_{0,o}) = 0, \quad \mathbb{E}\{(X_o - \mu_{0,o})(X_o - \mu_{0,o})'\} = \Sigma_{0,oo}.$$

Therefore the expected saturated normal score for that pattern is zero at $\eta_0 = (\mu_0, \text{vech}(\Sigma_0))$. Summing over the MCAR pattern distribution preserves this zero-score property. Under ordinary uniqueness conditions,

$$\eta^\dagger = \eta_0.$$

If the restricted SEM is also correct, so that

$$\eta(\theta_0) = \eta_0,$$

then the saturated curvature and the structured/model-implied curvature have the same asymptotic target:

$$H_\eta(\eta^\dagger) = H_\eta\{\eta(\theta_0)\}.$$

This statement does not require normality; it only uses the fact that, under MCAR, the observed-pattern normal score is unbiased at the full-data first two moments.

Conclusion. *For a correct model under MCAR, a persistent asymptotic gap between saturated FMG and lavaan-style MLR should not be explained by a different limiting definition of V . If the simulations show MCAR improvement for FMG over MLR, the first explanation should be the reference-law reduction: full spectrum versus scalar mean scaling.*

Where non-normality enters under MCAR

Under non-normality and MCAR, $\eta^\dagger$ can still equal the full-data mean and covariance, but J_η is not the normal information matrix. Consequently

$$\Gamma_{\text{mis}} = H_\eta^{-1} J_\eta H_\eta^{-1} \neq H_\eta^{-1},$$

and the nonzero eigenvalues of $U(V)\Gamma_{\text{mis}}$ need not be equal. This is the direct missing-data analogue of the complete-data ADF story [1, 10].

The Yuan–Bentler / MLR mean-scaled statistic uses only

$$c_{\text{SB}} = \frac{1}{\text{df}} \text{tr}\{U(V)\Gamma_{\text{mis}}\} = \frac{1}{\text{df}} \sum_{j=1}^{\text{df}} \lambda_j,$$

and compares T/c_{SB} to χ_{df}^2 . That is exact only in special cases, for example when the nonzero λ_j 's are all equal. It can be a poor approximation when the spectrum is dispersed.

This c_{SB} is precisely lavaan’s default `estimator = "MLR"` (Mplus `yuan.bentler.mplus`) scaling for incomplete data: the “MLR” branding hides that it is a *first-moment* correction, matching only the mean of the reference law. Its failure under a dispersed spectrum is therefore not a new phenomenon but the classical Satorra–Bentler mean-scaling limitation [7, 10] carried over to the FIML setting; the Mplus variant differs from the asymptotic trace only in finite-sample bookkeeping of $\text{tr}\{U(V)\Gamma_{\text{mis}}\}$, not in being mean-only. A dispersed spectrum is exactly what non-normality produces, so a mean scaling must over-reject whenever $\text{var}_j(\lambda_j)$ is large relative to $\bar{\lambda}^2$.

The FMG family keeps the spectrum. The exact all-eigenvalue calculation uses the distribution of $\sum_j \lambda_j \chi_{1j}^2$, which can be evaluated by standard quadratic-form methods such as Imhof’s method [5]. EBA and penalized FMG variants approximate or shrink the same eigenvalue spectrum [2, 3, 4]. Thus the MCAR headline comparison is not “different Gamma” and not necessarily “different V ”. It is usually

mean-only MLR scaling versus spectrum-aware FMG reference law.

What the Mplus variant actually computes

The previous section called c_{SB} lavaan’s `yuan.bentler.mplus` (the `estimator = "MLR"`) scaling and said the Mplus branding only hides a finite-sample bookkeeping of $\text{tr}\{U(V)\Gamma_{\text{mis}}\}$. This section makes that bookkeeping explicit, because it is the one place the Mplus default departs from the exact saturated-space construction used by FMG.

Write the saturated curvature and meat as $V = H_\eta$ and $J = J_\eta$, so that $\Gamma_{\text{mis}} = V^{-1}JV^{-1}$ and $U(V) = V - V\Delta(\Delta'V\Delta)^{-1}\Delta'V$. A cyclic trace splits the reference-law mean into two pieces,

$$\text{tr}\{U(V)\Gamma_{\text{mis}}\} = \underbrace{\text{tr}(V^{-1}J)}_{\text{h1: saturated}} - \underbrace{\text{tr}\{(\Delta'V\Delta)^{-1}(\Delta'J\Delta)\}}_{\text{h0: restricted}}.$$

Each term is an “effective number of parameters” $\text{tr}(A^{-1}B)$: the saturated H1 fit, minus the restricted SEM. Under information equality $J = V$ both traces collapse to ordinary parameter counts and the difference is exactly df; non-normality is what makes each $\text{tr}(A^{-1}B)$ depart from its parameter count.

The exact construction forms $U(V)$ and Γ_{mis} from V , J , and Δ directly. The Mplus variant instead reads the two traces off the two *separate* fits,

$$\text{tr}\{U(V)\Gamma_{\text{mis}}\} \approx \text{tr}(V_N^{-1}J_N) - \text{tr}(A_0^{-1}B_0),$$

where V_N, J_N are the saturated bread and meat at the EM fit $\hat{\eta}$ (with the H1 curvature forced *unstructured*; it is never read at the model-implied moments), A_0 is the restricted model’s own information, and $B_0 = \sum_i s_i s_i'$ is the cross-product of the restricted model’s casewise score vectors $s_i = \partial \ell_i / \partial \theta$. The remaining steps are unchanged: $c_{\text{SB}} = \text{tr} / \text{df}$, referred to χ_{df}^2 .

Where the slack is. The restricted meat is *exact*: by the score chain rule $s_i = \Delta' h_i$, so $B_0 = \Delta' (\sum_i h_i h_i') \Delta = \Delta' J_N \Delta$ identically. The approximation lives entirely in A_0 . The expected information satisfies $A_0 = \Delta' V \Delta$ exactly, because the expected (Gauss–Newton) curvature drops the second-derivative term. The observed Hessian keeps it:

$$A_0^{\text{obs}} = \Delta' V \Delta + \sum_k \frac{\partial^2 \eta_k}{\partial \theta \partial \theta'} g_k,$$

where g_k is the H1 moment gradient, proportional to the saturated-versus-model residual $\hat{\eta} - \eta(\hat{\theta})$. That extra term vanishes only at the population truth and as $N \rightarrow \infty$. So under the MLR observed-information default, $\text{tr}(A_0^{-1} B_0)$ only approximates $\text{tr}\{(\Delta' V \Delta)^{-1} \Delta' J \Delta\}$, and the two-trace difference equals $\text{tr}\{U(V) \Gamma_{\text{mis}}\}$ only up to that finite-sample curvature term [10, 8]. This is the precise content of “differs only in finite-sample bookkeeping, not in being mean-only.”

Two structural consequences. First, because the trace is read off two plug-in fits, the H1 curvature on this route is always the saturated/unstructured one; the structured V of the next section is not even an option for `yuan.bentler.mplus`. Second, the shortcut returns the trace but not $\text{tr}\{(U(V) \Gamma_{\text{mis}})^2\}$. With only the first spectral moment in hand, the Mplus variant cannot assemble a mean-and-variance-adjusted or scaled-shifted statistic from these matrices; it is *structurally* mean-only, however dispersed the λ_j actually are. The FMG family, which keeps J_N , V_N , and Δ and rebuilds the whole spectrum, is exactly what is unavailable once the trace has been collapsed this way.

Provenance. The mean-scaled FIML statistic and its exact trace are Yuan and Bentler’s [10]: their Eq. (18) writes the scaling as $\text{tr}(\hat{V}_{\hat{\beta}} \hat{V}_1)$ with $\hat{V}_1 = H_1 - H_1 \hat{\beta} (\hat{\beta}' H_1 \hat{\beta})^{-1} \hat{\beta}' H_1$, i.e. the projector $U(V)$ contracted with the sandwich covariance, generalizing Satorra–Bentler [7] to incomplete data. They document the *target*, not the shortcut. The two-trace realization above is the Mplus MLR implementation; the only thing that makes it Mplus rather than exact is the choice of A_0 : Mplus’s MLM uses the expected information

(for which $A_0 = \Delta'V\Delta$, and the shortcut is exact), while MLR uses the observed Hessian (for which it is only approximate). Mplus presents this as the robust correction, not as an approximation; the expected-versus-observed reading that exposes the slack is Savalei's [8]. The underlying decomposition $\text{tr}(A_1^{-1}B_1) - \text{tr}(A_0^{-1}B_0)$ is just the mean of a misspecified likelihood ratio [9], not specific to the FIML setting.

Structured versus saturated V

In words: the *saturated* (“unstructured”) choice reads the H1 curvature off the data alone, through the saturated EM moments $\hat{\eta}$; the *structured* choice reads it off the fitted SEM, through the model-implied moments $\eta(\hat{\theta})$. They share a probability limit under a correct MCAR null, so the experiment's two `_structured` columns are a finite-sample / robustness check, not a different estimand. They separate only under misspecification or non-normal MAR, as below.

The structured H1 analogue evaluates the same normal H1 curvature at the restricted fitted moments,

$$V_{\text{struct}} = H_{\eta}\{\eta(\hat{\theta})\},$$

whereas saturated FMG evaluates it at the saturated H1 fit,

$$V_{\text{sat}} = H_{\eta}(\hat{\eta}).$$

If both constructions are forced into the same eta-space comparison with the same Γ_{mis} , then the only algebraic difference is

$$\text{eig}\{U(V_{\text{struct}})\Gamma_{\text{mis}}\} \quad \text{versus} \quad \text{eig}\{U(V_{\text{sat}})\Gamma_{\text{mis}}\}.$$

Under the correct MCAR null these two V 's share the same probability limit, but their finite-sample traces can differ because one uses $\hat{\theta}$ -implied moments and the other uses the saturated EM moments. The advisory trace-convergence check in this repository is exactly about that finite-sample distinction.

The V -evaluation distinction matters asymptotically in other regimes:

1. under restricted-model misspecification, because $\eta(\theta^*) \neq \eta^\dagger$;
2. under non-normal MAR working likelihoods, because the observed-data normal quasi-likelihood target can differ from the full-data moment target.

The second case is not the clean MCAR calibration problem. Normal FIML has its usual likelihood justification under MAR when the observed-data normal model is correctly specified. Under non-normal MAR, Gaussian FIML is a working likelihood; it may be biased for the full-data moment model, and the LRT can be noncentral under the intended SEM null. No reference-law correction can fully fix that estimator-target problem [10, 8].

Two-variable V witness

The following complete-data example is useful only as a witness for the V_{struct} versus V_{sat} distinction. It is not the MCAR null mechanism.

Let the saturated covariance of two observed variables be

$$\Sigma_Y = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}, \quad \rho \neq 0,$$

but fit the restricted independence model

$$\Sigma(\theta) = \begin{bmatrix} \theta_1 & 0 \\ 0 & \theta_2 \end{bmatrix}.$$

Then the restricted pseudo-true covariance is I_2 , while the saturated covariance is Σ_Y . The gap is misspecification; it exists even under normal data.

Use covariance moments ordered as

$$\eta = (\sigma_{11}, \sigma_{21}, \sigma_{22})'.$$

The restricted derivative is

$$\Delta = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

At the structured point I_2 ,

$$\Gamma_{\text{NT}}(I_2) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad V_{\text{struct}} = \Gamma_{\text{NT}}(I_2)^{-1},$$

and

$$U_{\text{struct}} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

At the saturated covariance,

$$\Gamma_{\text{NT}}(\Sigma_Y) = \begin{bmatrix} 2 & 2\rho & 2\rho^2 \\ 2\rho & 1 + \rho^2 & 2\rho \\ 2\rho^2 & 2\rho & 2 \end{bmatrix}.$$

If the trace uses U_{struct} with the saturated covariance-moment matrix, then

$$c_{\text{struct}} = \text{tr}\{U_{\text{struct}}\Gamma_{\text{NT}}(\Sigma_Y)\} = 1 + \rho^2.$$

By contrast, the unstructured/saturated choice uses

$$V_{\text{sat}} = \Gamma_{\text{NT}}(\Sigma_Y)^{-1},$$

and $U(V_{\text{sat}})\Gamma_{\text{NT}}(\Sigma_Y)$ is a rank-one projector, so

$$c_{\text{sat}} = 1.$$

This proves that the evaluation point of V can matter. It does not prove that V is the explanation for non-normal MCAR null calibration failures.

The MAR mechanism used (and why it is weak)

MAR is not one mechanism, and the strength of the estimator-target problem above depends entirely on which MAR is used. Experiment 23 uses the Savalei–Bentler 2005 MAR generator [8]: two indicators x_1, x_2 are always observed and act as the missingness cause, and the remaining indicators $x_3, \dots, x_6$ are deleted by coarse *sign* rules on the always-observed pair (rows with $x_1 + x_2 > 0$, and rows with $x_1 < 0$), with the deletion probabilities calibrated to the target overall rate. This is a deliberately mild MAR for two reasons:

1. the selection depends only on the *sign* of always-observed indicators, a binary threshold rather than a steep continuous function, so the conditional missingness probability takes only a few levels; and
2. because x_1, x_2 enter the FIML likelihood, Gaussian FIML recovers the conditional mean of the missing entries given the observed ones *exactly* whenever that conditional mean is linear, and the Vale–Maurelli / inverse-Gaussian / power-law generators keep conditional means close to linear. The non-normality lives in the marginals, not in a curved regression surface.

Consequently the estimator-target bias that the structured-versus-saturated section flags as the irreducible MAR limitation is *small* in this design, and the MAR cells behave almost like MCAR for the well-calibrated tests. This is a property of *this* MAR, not of MAR in general: a stronger mechanism (steep continuous selection on the observed scores, or a generator with a genuinely nonlinear missing-given-observed conditional mean) would make $\eta^\dagger \neq \eta_0$ bite, push the base statistic non-central under the intended SEM null, and leave a residual no reference-law correction can remove.

Size-adjusted power

Ordinary power rewards over-rejection: a test that rejects 27% of *true* models also rejects more false ones, but that detection is borrowed from a broken Type-I rate. Size-adjusted (empirical-size-corrected) power removes the loan. For a fixed cell and method, let F_0 be the simulated distribution of the p -value under the true model, and set the critical threshold to its α -quantile,

$$c = F_0^{-1}(\alpha),$$

the cutoff that *would* give exactly size α in that cell. The size-adjusted power is then $P(p < c)$ under the alternative. A well-calibrated test has $c \approx \alpha$ and ordinary $\approx$ size-adjusted power; a liberal test has $c < \alpha$, which is exactly how its inflated raw power is clawed back. This is a diagnostic, not an operational procedure: c is read from the null simulation and is unavailable to a practitioner with one data set. Its purpose is to compare the *intrinsic* discriminating ability of the statistics on equal honest footing.

Diagnostics for experiment 23

For the MCAR cells, the clean diagnostic sequence is:

1. Compare lavaan MLR with magmaan's FMG *SB* row. Both are mean-only trace scalings. If they agree and both miscalibrate, the culprit is the scalar reference-law approximation.
2. Inspect the eigenvalue dispersion of UT_{mis} . Large spread predicts poor mean-scaling performance and gives FMG room to improve.
3. Check that $\sum_j \lambda_j$ from saturated FMG approaches the FIML MLR trace as N grows under MCAR. A persistent large gap points to a convention or implementation mismatch in V , Δ , or scaling.

4. Treat MAR cells separately. Mean base statistic exceeding mean trace is an estimator-target/noncentrality warning, not a failure of the eigenvalue reference law.

The `noncentrality.csv` diagnostic reports, per h_0 cell,

$$\widehat{\text{ncp}} = \overline{T} - \overline{\text{tr}\{U(V)\hat{\Gamma}_{\text{mis}}\}} = \text{mean_base} - \text{mean_trace},$$

the gap between the mean base LRT and the mean of its own reference law. With a consistent estimator under a correct null $\mathbb{E}[T] \rightarrow \text{tr}\{U(V)\Gamma_{\text{mis}}\}$, so $\widehat{\text{ncp}} \rightarrow 0$; a positive value is the estimated non-centrality the base statistic carries *beyond* what mean scaling can absorb. One caveat reading it: in finite samples the sandwich trace $\text{tr}\{U(V)\hat{\Gamma}_{\text{mis}}\}$ is itself downward biased under heavy non-normality, so a moderate positive $\widehat{\text{ncp}}$ under *MCAR* is largely a finite-sample sandwich-bias artifact and shrinks with N ; it does not contradict the calibration of the spectrum-aware tests, which match the same finite-sample spectrum. Only a positive $\widehat{\text{ncp}}$ that *persists* as N grows – the strong non-normal MAR case – signals the estimator-target bias that no reference-law correction removes.

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